

Chow

Technical / water-system voice

How to use this: not a verbatim script — nobody is expected to read from it word-for-word, and realistically no one will stick to a script anyway. Study it well enough to explain each point in your own words, improvise if you're cut off, and answer a follow-up without freezing. If a question lands outside your section, give the short version and hand off by name — every presenter's script and the underlying `FIELD-NOTES.md` are in the same GitHub repo.

You're the technical voice — the Batbat numbers and the field tools are both yours, so you carry the deck's hardest data slide and its most concrete engineering slide back to back with only Ash's callout and Nancy's farm photos between them. The rest of the team doesn't need to go deep on this material.

Slide 8 — The numbers from Batbat (3 charts).

1. *Loss by cause: cacao vs. rice vs. dragon fruit* — a grouped bar chart. Each cause has three bars: our cacao numbers next to two other Philippine cash crops. - **Handling:** cacao loses **60%** of pods to careless handling (field notes); Philippine rice loses **16.5%** total post-harvest [5]; dragon fruit loses **35%** of output post-harvest [17]. - **Pests & disease:** cacao **80–100%** (bar to 80, whisker to 100 — say the range out loud); rice **24–41%**, average ~37% of yield (IRRI, tropical Asia) [18]; dragon fruit stem canker **60–80%** in severe cases [19]. - **Typhoon (bad year):** cacao ~**100%**; rice **up to 95%** of standing crop in the worst-hit regions post-Haiyan [20]; dragon fruit **100%** — growers report total production loss when a typhoon topples the posts the vines grow on (storm Mario: 300 posts, ~4 t) [21]. If pressed on the dragon fruit figure, be honest that it's grower-reported rather than a published statistic. - The takeaway to land: even against crops with famously bad loss profiles, **cacao's pest/disease bar is the outlier** — worse than rice's worst case and above dragon fruit's most destructive disease.
2. *Tatay's cacao yield:* the chart shows all three bars — **500 kg/year** in previous years, near **0 kg** this season due to pest/disease — his estimated potential if pests were resolved is **2,000 kg/year**. Say this slowly: that's a **4× recovery**, and it's the single largest lever in the entire dataset.
3. *Farmer monthly income benchmarks:* a ₱2,000 low reference, **Blandino Papina** at ₱5,000 as the named example farmer, and ₱10,000 named as a "low" ceiling. The dashed line is Bicol's **poverty threshold for a family of five: ₱13,624/month** [4] — even the ₱10,000 "ceiling" sits ₱3,600 below it. Context for how thin margins are region-wide, not just for Tatay. The footnote adds broad context: PH fruits and vegetables lose 15–35% post-harvest [6], and cacao is far beyond that.

Slide 11 — Run the pilot through a live, calculated tool.

Explain what you built and why: rather than eyeballing a drip-irrigation pilot for Tatay's farm, you built a working set of field tools —

- a **drip-pressure planner** sized to his real 2.5-hectare / 120-meter layout,
- a **Raspberry Pi controller schematic** — the automation block that opens the zone valves (and the pump) on a schedule set by weather and season, from soil-moisture, rain, temperature and tank-level sensors, and
- an **interactive schematic editor**

(There's also a full **water-system schematic** — gravity vs. pump modes — reachable from the homepage and cross-linked from the controller page; it isn't a card on the slide, but mention it if someone asks how the pieces connect.)

— so any water-resilience pilot for Tatay is calculated from his actual numbers, not guessed. The water math itself is cited on the slide ([3]–[8]: the DA cacao production guide, the cocoa water-relations literature, FAO-56). Point the room at the GitHub repo (github.com/Abdullah-Nasir-Chowdhury/cacao-bicol-field-tools), which also holds the full field-notes reference and the interactive chart dashboard this deck draws its numbers from. Keep the framing tight: this is the concrete, checkable, technical contribution — invite people to open it later rather than trying to demo it live if time is short.